4. Implement House price Prediction using appropriate machine learning algorithm.

CODE:

```
# House Price Prediction using Linear Regression
# Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
# Load California Housing Dataset
housing = fetch_california_housing()
# Features and Target
X = housing.data
y = housing.target
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# Create Linear Regression Model
model = LinearRegression()
# Train Model
model.fit(X_train, y_train)
# Predict House Prices
y_pred = model.predict(X_test)
# Model Evaluation
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
print("House Price Prediction using Linear Regression")
print("---------------------------------------------")
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
print("R2 Score:", r2)
```

```
[ ]
# Actual vs Predicted Values
result = pd.DataFrame({
    "Actual Price": y_test[:10],
    "Predicted Price": y_pred[:10]
})
print("\nSample Predictions")
print(result)
# Predict Price for One House
sample_house = X_test[0].reshape(1, -1)
predicted_price = model.predict(sample_house)
print("\nPredicted House Price:", predicted_price[0])
# Plot Actual vs Predicted Prices
plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred, color="blue")
plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    color="red",
    linewidth=2
)
plt.title("Actual vs Predicted House Prices")
plt.xlabel("Actual House Price")
plt.ylabel("Predicted House Price")
plt.grid(True)
plt.show()
```

OUTPUT:

```
House Price Prediction using Linear Regression
---------------------------------------------
Mean Squared Error (MSE): 0.5558915986952422
Root Mean Squared Error (RMSE): 0.7455813830127749
R2 Score: 0.5757877060324524

Sample Predictions
   Actual Price  Predicted Price
0       0.47700         0.719123
1       0.45800         1.764017
2       5.00001         2.709659
3       2.18600         2.838926
4       2.78000         2.604657
5       1.58700         2.011754
6       1.98200         2.645500
7       1.57500         2.168755
8       3.40000         2.740746
9       4.46600         3.915615

Predicted House Price: 0.7191228416021715
```